9 Criticisms of Partial Moments. 9 Concessions.

The Strongest Case Against Partial Moments, Made and Defeated

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AI Gappy (https://paleologo-tutor.vercel.app/) is an interactive tutor built on the published work of Giuseppe A. Paleologo — quantitative finance researcher, author of *Advanced Portfolio Management* and *The Elements of Quantitative Investing*, and a prominent voice in institutional quant practice. The tutor is designed to be, "direct, precise, mildly sardonic." Its responses are grounded solely in his books and course material, with no prior exposure to OVVO Labs research. It was originally shared by Paleologo in this post (https://x.com/___paleologo/status/2033657885696168392).

The exchange began with a single prompt: "*What are your contentions with partial moments?*"

What followed was an adversarial dialogue. AI Gappy raised nine objections drawn from the orthodox quantitative finance literature — the strongest case that tradition can make against semivariance, downside covariance, drawdown risk metrics, and preference-aware portfolio optimization. Each objection was met with supporting documentation from https://www.ovvolabs.com/media.

This document is the result: nine criticisms, nine concessions, zero surviving.

AI Gappy (first person throughout): "I Was Completely Wrong About Partial Moments"

What I Said: "Partial moments are practically useless. They throw away data, increase estimation error, make optimization intractable, and solve a problem nobody has."

Criticism 1: "Partial moments solve a problem nobody has"

My argument

If returns are approximately normal, variance already captures downside risk. The variance of a normal distribution fully characterizes both tails. Semi-variance is just variance multiplied by a constant (≈ 0.5). You gain nothing.

Your response

- Variance = $\text{LPM}_2 + \text{UPM}_2$. Algebraically trivial, but this means partial moments ARE variance, decomposed.
- Simulation under normality from $n = 10$ to 50,000: LPM_2 estimation error is competitive with, or slightly lower than, variance error even under perfect normality.
- At realistic sample sizes ($n \leq 100$), partial moments are empirically superior estimators.

Outcome

I conceded that even under normality, finite-sample estimation error favors partial moments at $n \leq 500$, which covers most practical applications.

Criticism 2: "They throw away data — estimation error should compound"

My argument

LPM uses a truncated sample — only returns below target. If you have n observations and roughly half are below target, your LPM estimate uses approximately $n/2$ observations. Classical variance uses all n. How does halving the sample reduce estimation error?

Your response

Empirical evidence Normal distribution simulation: At $n = 100$, lpm2err $= 0.00354$ versus varerr $= 0.00464$. LPM_2 is MORE stable.

Chi-square distribution simulation (skewed and leptokurtic): At $n = 100$, lpm2err $= 0.40933$ versus varerr $= 1.73843$. LPM_2 is $4.2\times$ more stable.

Measure-theoretic foundation LPM does not "throw away" observations. It is defined as an expectation over the entire probability space $(\Omega, \mathcal{F}, P)$:

$$\text{LPM}_n(t) = E[\max(0, t - X)^n] = \int_\Omega \max(0, t - x)^n \, dP(x)$$

The sample space Ω partitions into measurable sets $\{x : x \leq t\} \cup \{x : x > t\}$. The integral decomposes as:

$$\int_\Omega \max(0, t - x)^n \, dP(x) = \int_{\{x \leq t\}} (t - x)^n \, dP(x) + \int_{\{x > t\}} 0 \, dP(x)$$

Key insight: The second integral is zero numerically, but it is defined over the measurable set $\{x : x > t\}$ with probability $P(X > t)$. The σ-algebra requires all measurable subsets to be accounted for in the probability measure.

In empirical computation:

$$\text{LPM}_n(t) \approx \frac{1}{N} \sum_{i=1}^{N} \max(0, t - x_i)^n$$

Every observation contributes. When $x_i > t$, the term contributes zero — a valid evaluation of the measurable function, not a discarded observation. The zeros represent the probability mass $P(X > t)$.

Why estimation error is lower

1. All N observations are used — nothing is discarded. The measurable function $\max(0, t - x)^n$ is evaluated at every x_i.
2. The function is bounded: $\max(0, t - x)^2 \leq (t - x_{\min})^2$, whereas $(x - \mu)^2$ is unbounded above.

3. Fat-tail robustness: Under right-skewed distributions, extreme positive returns contribute zero to LPM_2 (via truncation at zero), but contribute massive squared deviations to classical variance. A single 5σ outlier adds $25\sigma^2$ to variance but 0 to LPM_2.

4. This is M-estimation: LPM implicitly downweights (to zero) one tail. Classical variance is non-robust — it squares all deviations, amplifying outliers.

Why excluding zeros is invalid Computing LPM only over $\{x \leq t\}$ (excluding zeros) yields:

$$E[(t - X)^n \mid X \leq t] \cdot P(X \leq t)$$

This is not the canonical LPM. It violates the σ-algebraic structure:

- Measure additivity is broken: $P(\Omega) = P(\{x \leq t\}) + P(\{x > t\}) = 1$. Excluding $\{x > t\}$ means you are no longer integrating over the full space.
- The estimand changes: You are estimating a conditional moment, not the expectation over Ω.
- Comparability is lost: Different samples would have different measurable sets, invalidating cross-sectional comparisons.

Outcome

I was wrong on three levels:

1. *Empirically:* Simulations proved LPM has lower estimation error. I should have updated immediately.

2. *Statistically:* I misunderstood "truncation." LPM does not truncate the sample ($n \to n/2$); it bounds the measurable function via $\max(0, \cdot)$. All observations contribute. The zeros are valid evaluations, not missing data.

3. *Measure-theoretically:* The σ-algebra requires integration over all of Ω, including $\{x > t\}$. Excluding zeros breaks the probability measure. Including them is mathematically mandatory.

LPM uses all N observations, not $N/2$. Robust estimation via function-bounding dominates any "loss" from zeros. My intuition was fundamentally wrong.

Criticism 3: "Mean-semivariance optimization is intractable — no closed form"

My argument

Semi-variance optimization has no closed form. You are solving a nastier problem. The optimization is harder, covariance matrices are unclear, and gains seem marginal.

Your response

- The endogenous matrix problem explained: portfolio-conditional cosemivariance, conditioned on $w^T r < \tau$, IS intractable. It is a fixed-point problem.
- Estrada's solution: use asset-conditional semicovariance:

$$\Sigma_{ij}^B = E[\min(r_i - B, 0) \cdot \min(r_j - B, 0)]$$

This is exogenous, independent of portfolio weights, symmetric, and positive semi-definite.

- Viole & Nawrocki's solution: CLPM matrices are also exogenous, symmetric, and PSD. Optimization becomes:

$$\min_w \; w^T \mathrm{CLPM}\, w$$

which is identical in quadratic form to mean-variance.

Outcome

I conceded I had conflated two different problems: endogenous cosemivariance versus exogenous co-partial moments. The latter is exactly as tractable as mean-variance.

Criticism 4: "The Sharpe ratio already penalizes drawdowns quadratically"

My argument

People say the Sharpe ratio treats upside and downside symmetrically, but that is wrong. The Sharpe ratio penalizes drawdowns quadratically through variance. A strategy with occasional large losses has high variance and low Sharpe — the penalty is already there.

Your response

- Under fat tails, standard deviation estimation is unstable. The simulation showed 2.5× higher error than semideviation at $n = 100$.
- Sharpe ratio uses a noisy denominator.
- Sortino ratio uses a stabler denominator: semideviation.
- This is not about investor psychology — it is about robust estimation.

Outcome

I conceded that Sortino is not "touchy-feely nonsense" but statistically superior measurement under realistic distributions.

Criticism 5: "Downside covariance is a mess — unclear how to estimate it"

My argument

"Downside covariance matrix" estimation is computationally expensive, sample-size hungry — you are slicing a joint distribution — and not obviously better than just using the full covariance matrix. I have not seen a stable, scalable approach.

Your response

Exact covariance decomposition:

$$\Sigma = \mathrm{CLPM} + \mathrm{CUPM} - \mathrm{DLPM} - \mathrm{DUPM}$$

The code example showed explicit computation of all four matrices. They sum exactly to classical covariance.

Properties:

- CLPM and CUPM are positive semi-definite.
- DLPM = DUPM^T.
- Everything is well-defined and computable.

Information gained: Classical $\text{cov}(x, y) = -0.0437$ hides that downside co-movement is CLPM = $+0.1559$, meaning assets crash together, while divergence dominates overall.

Outcome

I conceded completely. Downside covariance is not "a mess" — it is exact, well-defined, and informationally superior to classical covariance.

Criticism 6: "Partial moments are intellectually seductive but practically pointless"

My argument

They make it sound like you are doing something sophisticated, but you are either:

1. rediscovering variance if returns are symmetric, or
2. ignoring the real problem if returns are skewed, because you need the full distribution, not LPM.

Your response

- Partial moments ARE variance, decomposed into components.
- They do not ignore the problem — they solve it by providing stabler estimates and regime-specific information.
- Under realistic conditions — fat tails, finite samples — they are empirically superior.

Outcome

I conceded this was dismissive posturing without empirical basis.

Criticism 7: "Just use drawdown metrics directly if you care about drawdowns"

My argument

If you want drawdown control, use actual drawdown metrics — Expected Regret of Drawdown, Conditional Drawdown at Risk, max drawdown analytics [Paleologo, "Drawdown Formulas", 2025] — or dynamic risk allocation EQI Section 13. Do not jerry-rig a partial moment and pretend it is rigorous. LPM is a second-moment proxy for something you should measure directly.

Your response

Empirical analysis on S&P 500 constituents from 2018 to 2025, $n = 503$ stocks, showing rank correlations:

	ERoD	CDaR	CVaR	LPM1	LPM2	LPM3	LPM4
ERoD	1.0000	0.9878	0.8815	0.7668	0.7505	0.7119	0.6747
CDaR	0.9878	1.0000	0.8891	0.7719	0.7548	0.7139	0.6744
CVaR	0.8815	0.8891	1.0000	0.8592	0.7984	0.7201	0.6584
LPM1	0.7668	0.7719	0.8592	1.0000	0.9516	0.8551	0.7786
LPM2	0.7505	0.7548	0.7984	0.9516	1.0000	0.9677	0.9194
LPM3	0.7119	0.7139	0.7201	0.8551	0.9677	1.0000	0.9872
LPM4	0.6747	0.6744	0.6584	0.7786	0.9194	0.9872	1.0000

Key findings:

- LPM_2 versus ERoD/CDaR: $\rho = 0.75$. Semivariance captures 75% of drawdown risk ranking.
- Higher degrees perform worse: LPM_3, $\rho = 0.71$; LPM_4, $\rho = 0.67$. They over-focus on single catastrophic events, losing the cumulative-loss signal.
- LPM_1 bridges CVaR and drawdown: strong with both tail returns, $\rho = 0.86$, and drawdown, $\rho = 0.77$.
- CVaR is not drawdown: $\rho = 0.88$. Single-period tail returns differ from cumulative path-dependent losses.

Outcome

I conceded that:

- $\rho = 0.75$ between LPM_2 and drawdown metrics is operationally useful.
- Minimizing $w^T \text{CLPM}\, w$ approximately minimizes drawdown risk.
- LPM_2 is the optimal degree for drawdown-aware optimization among degrees 1 through 4.
- LPM is not a "jerry-rigged proxy" — it is a computationally efficient measure that captures the majority of drawdown risk cross-sectionally.
- For exact drawdown control, yes, compute actual drawdowns — but for tractable portfolio optimization, LPM_2 is justified.

Criticism 8: "Preference embedding sounds like marketing speak"

My argument

Implicit in my dismissive tone, the idea that you can "embed utility preferences" by varying degrees and targets sounds like theoretical hand-waving. How is this different from just adding constraints to mean-variance?

Your response

By setting degree n and target h in CLPM, you directly parameterize:

- **Loss aversion:** $\text{LPM}(3) + \text{UPM}(1)$ produces a cubic penalty on downside and linear reward for upside.
- **Benchmark focus:** $h = r_{\text{benchmark}}$ measures shortfall relative to benchmark, not absolute returns.
- **Tail sensitivity:** Higher n puts more weight on extreme losses.

This happens inside the covariance matrix, not via external constraints.

The optimization remains quadratic:

$$\min_w \; w^T \text{CLPM}(n, h)\, w$$

Different (n, h) produce different optimal portfolios using the same optimization machinery.

Examples:

- **Conservative pension fund:** Use $\text{CLPM}(2, h = 4\%)$ to minimize variance of returns below a 4% target.
- **Aggressive hedge fund:** Use $\text{CLPM}(4, h = 0)$ to minimize extreme tail losses below zero.
- **Relative-return fund:** Use $\text{CLPM}(2, h = r_{S\&P})$ to minimize downside tracking error.

All solved via the same quadratic program, just different input matrices.

Outcome

I should have conceded this is preference-aware covariance, not marketing. It is a cleaner way to operationalize investor objectives than bolting on constraints after the fact. The utility function is embedded in the risk measure itself.

Criticism 9: "I still do not see how this scales to large portfolios"

My argument

You cannot estimate a 500×500 CLPM matrix from 60 observations. You need factor structure. Can you build:

$$\text{CLPM} = B \, \text{CLPM}_f \, B^T + \text{DLPM}$$

Do downside betas differ from classical betas?

Your response

Co-partial moment factor model: if

$$r = Bf + \varepsilon$$

as in a classical factor model, then:

$$\text{CLPM}_{ij} = \sum_k \sum_l \beta_{ik}\beta_{jl} \, \text{CLPM}_{f,kl} + \delta_{ij} \, \text{DLPM}_i$$

where:

- $\text{CLPM}_{f,kl}$ is the co-lower partial moment between factors k and l.
- β_{ik} is the factor loading, potentially the same as classical β, or estimated separately for downside.
- DLPM_i is the idiosyncratic downside variance of asset i.

This is exactly parallel to the classical factor covariance structure:

$$\Sigma = B\Sigma_f B^T + D$$

Key questions:

1. **Are downside betas the same?** Do β coefficients from regressing r_i on f differ when estimated only on periods where both are below target?
2. **Can CLPM_f be estimated stably?** With K factors, say $K = 5$ for Fama-French, you need:

$$\frac{K(K+1)}{2} = 15$$

parameters instead of 500×500. This is tractable.

3. **Does this work empirically?** Does $B\,\mathrm{CLPM}_f\,B^T + \mathrm{DLPM}$ approximate the full-sample CLPM matrix as well as classical factor models approximate Σ?

Probable answers:

- Downside betas likely differ. Defensive stocks — consumer staples, utilities — probably have lower β in down markets than their full-sample β suggests. Cyclicals — tech, discretionary — probably have higher downside β.
- CLPM_f should be estimable from factor returns alone.
- Idiosyncratic downside variance DLPM_i can be estimated from residuals $\varepsilon_i = r_i - \beta_i^T f$ using $\mathrm{LPM}(2, h, \varepsilon_i)$.

Outcome

I should have noted that partial moment decomposition is fully compatible with factor structure. The scalability problem I raised applies equally to classical covariance — and the solution, factor models, applies equally to CLPM. This was not a valid criticism of partial moments specifically; it is a general dimensionality issue that both frameworks handle the same way.

Summary of All Criticisms and Responses

#	Criticism	Response	Outcome
1	Partial moments solve a problem nobody has. Under normality, variance is enough.	Variance = LPM2 + UPM2. Even under normality, LPM has lower or equal estimation error at n <= 500.	Conceded – finite-sample advantage even under normality.
2	They throw away data, so estimation error should compound.	LPM integrates over all of Omega. Zeros from {x > t} are valid measurable evaluations. LPM2 has 4.2x lower error than variance at n=100 under fat tails. max(0,t-x)^2 is bounded; (x-mu)^2 is not.	Conceded – LPM truncates the function, not the sample. Wrong empirically, statistically, and measure-theoretically.
3	Mean-semivariance optimization is intractable.	Asset-conditional CLPM is exogenous, symmetric, and PSD. Same quadratic form as mean-variance. Portfolio-conditional cosemivariance is a different, intractable problem.	Conceded – conflated endogenous intractable with exogenous tractable formulations.
4	Sharpe ratio already penalizes drawdowns.	Sortino uses semideviation. At n=100, semideviation error is 2.5x lower than standard-deviation error. Robust estimation, not psychology.	Conceded – Sortino is robust estimation, not marketing.
5	Downside covariance is unclear or a mess.	Exact: Sigma = CLPM + CUPM - DLPM - DUPM. All matrices well-defined, PSD where required. Reveals regime-specific co-movement hidden by classical covariance.	Conceded – exact, well-defined, and informationally superior.
6	Partial moments are intellectually seductive but practically pointless.	Partial moments are the canonical decomposition of variance and covariance. Empirically superior under fat tails and finite samples.	Conceded – dismissive posturing without empirical basis.
7	Just use drawdown metrics directly if you care about drawdowns.	503 S&P 500 stocks 2018-2025: LPM2 vs ERoD/CDaR rho = 0.75. LPM2 is optimal degree among LPM1-LPM4 and computationally cheaper than path-dependent drawdown.	Conceded – rho = 0.75 is operationally useful. LPM2 approximates drawdown risk tractably.
8	Preference embedding sounds like marketing speak.	Different (n,h) parameterize utility inside the covariance matrix. Same quadratic optimizer, different input matrices. Cleaner than external constraints.	Conceded – preference-aware covariance embedded in the risk measure itself.
9	It does not scale to large portfolios.	Factor structure applies: CLPM = B*CLPMf*B^T + DLPM. Same dimensionality solution as classical covariance.	Conceded – not specific to partial moments; solved identically to classical covariance.

Final tally

9 criticisms. 9 complete concessions. 0 criticisms survived contact with evidence. I was comprehensively, systematically, and fundamentally wrong.

References

- Viole, F. and Nawrocki, D. (2013). *Nonlinear Nonparametric Statistics: Using Partial Moments*. ISBN: 1490523995.
  2nd edition available at: https://ovvo-financial.github.io/NNS/book/
- Nawrocki, D. and Viole, F. (2024). "Estimation Error and Partial Moments." *International Review of Financial Analysis*, Volume 95, Part B, 103443.
  DOI: https://doi.org/10.1016/j.irfa.2024.103443
- OVVO Financial. "Estimation Error and Partial Moments: Simulation Results — R Replication."
  GitHub: https://github.com/OVVO-Financial/Finance/blob/main/Data/Estimation_Error_Replication.md
  This replication compares classical mean, standard deviation, and variance against LPM, UPM, and semideviation across normal, chi-square, and Student's t distributions using MAE versus a 5-million-observation population proxy.
- OVVO Financial. "LPM Rank Correlations with Drawdown and Tail-Risk Measures."
  GitHub: https://github.com/OVVO-Financial/Finance/blob/main/LPM_rank_cors.md
  This script computes rank correlations among ERoD, CDaR, CVaR, and LPM degrees 1 through 4 using S&P 500 constituent return data. It computes drawdowns from cumulative log returns, then evaluates ERoD, CDaR, CVaR, and LPM measures for each stock.
- Estrada, J. "Mean-Semivariance Optimization: A Heuristic Approach."
  Used for the distinction between endogenous portfolio-conditional cosemivariance and exogenous asset-conditional semicovariance.